

ORIGINAL ARTICLE

Left Atrial Appendage Closure Versus Oral Anticoagulation in Atrial Fibrillation: A Systematic Review and Meta-Analysis of Randomized Controlled Trials

Adishwar Rao¹ MD, MPH; Venkatesh Ravi¹ MD; Mukunthan Murthi¹ MD; Rahul Chikattimalla¹ MD; Vennela Talla¹ MBBS; Nihal Nagaraj¹ MBBS; Gaurav Dhar¹ MD; Neeraj Jolly, MD; Aviral Vij¹ MD, MS

BACKGROUND: Left atrial appendage closure (LAAC) is an alternative to oral anticoagulation (OAC) for stroke prevention in patients with atrial fibrillation, but randomized controlled trials have primarily focused on composite end points, potentially obscuring differences in individual clinical outcomes. We compared the efficacy and safety of LAAC versus OAC with an emphasis on individual clinical outcomes.

METHODS: A systematic review and meta-analysis of randomized controlled trials comparing LAAC with OAC in adults with atrial fibrillation was conducted in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses guidelines. Databases were searched from January 2000 to March 2026. The primary efficacy end point was any stroke, and the primary safety end point was major bleeding. Risk ratios (RRs) with 95% CIs were pooled using a random-effects model.

RESULTS: Six randomized controlled trials comprising 7004 patients (3681 LAAC; 3323 OAC) were included. LAAC was associated with a higher risk of ischemic stroke (RR, 1.41 [95% CI, 1.04–1.91]; $P=0.03$; 3.2% versus 2.1%), without a significant difference in any stroke (RR, 1.08 [95% CI, 0.82–1.42]; $P=0.60$) or the composite of stroke, systemic embolism, or death (RR, 0.99 [95% CI, 0.77–1.29]; $P=0.96$). Cardiovascular mortality (RR, 0.88; $P=0.40$) and all-cause mortality (RR, 0.92; $P=0.36$) were similar between groups. Major bleeding, including procedural events, did not differ significantly (RR, 0.93; $P=0.40$), although LAAC was associated with a significantly lower risk of nonprocedural major bleeding (RR, 0.57 [95% CI, 0.43–0.77]; $P<0.01$; 6.2% versus 10.7%).

CONCLUSIONS: In patients with atrial fibrillation, LAAC is associated with a higher risk of ischemic stroke, but lower risk of nonprocedural major bleeding compared with OAC, with no significant differences in composite thromboembolic outcomes or mortality, underscoring the importance of individualized risk-benefit assessment in stroke prevention strategies.

GRAPHIC ABSTRACT: A graphic abstract is available for this article.

Key Words: anticoagulants ■ atrial fibrillation ■ hemorrhage ■ ischemic stroke ■ left atrial appendage closure

Atrial fibrillation (AF) is the most common sustained cardiac arrhythmia in adults and is associated with an increased risk of ischemic stroke and mortality.¹ Oral anticoagulants (OACs) are the cornerstone of stroke prevention, with direct OACs (DOACs) favored over vitamin K antagonists (VKAs).^{2,3} However, bleeding

complications remain a major limitation, with annual major bleeding rates of 2.1% to 3.6%, fatal bleeding rates of up to 0.5%, and discontinuation of DOAC therapy in 20% to 30% of patients within 2 years.^{4,5} Therefore, left atrial appendage closure (LAAC) or LAA occlusion devices are used as a mechanical alternative to pharmacological

Correspondence to: Venkatesh Ravi, MD, Division of Electrophysiology, Heart and Vascular Institute, Saint Francis Health System, 6151 S Yale Ave, Ste A100, Tulsa, OK 74136. Email venkateshravi.8711@gmail.com

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WHAT IS KNOWN?

- Oral anticoagulation is the standard therapy for stroke prevention in atrial fibrillation, whereas left atrial appendage closure offers a mechanical alternative for selected patients who cannot tolerate or prefer to avoid long-term anticoagulation.
- Randomized trials of left atrial appendage closure have largely emphasized composite efficacy and safety outcomes, which may obscure clinically important differences in individual ischemic and bleeding events.

WHAT THE STUDY ADDS

- In a meta-analysis of 6 randomized trials, including 7004 patients, left atrial appendage closure was associated with a higher risk of ischemic stroke than oral anticoagulation, despite no significant difference in any stroke, the composite of stroke, systemic embolism, or death, or mortality.
- Left atrial appendage closure reduced nonprocedural major bleeding, but major bleeding was similar when procedural events were included. These findings highlight the trade-off between ischemic and bleeding risks and support individualized treatment selection and shared decision-making.

Nonstandard Abbreviations and Acronyms

AF	atrial fibrillation
CRNMB	clinically relevant nonmajor bleeding
DOAC	direct oral anticoagulant
HAS-BLED	Hypertension, Abnormal Renal/Liver Function, Stroke, Bleeding History or Predisposition, Labile International Normalized Ratio, Elderly, Drugs/Alcohol
LAAC	left atrial appendage closure
LAAOS-4	The Fourth Left Atrial Appendage Occlusion Study
OAC	oral anticoagulation
RCT	randomized controlled trial
RR	risk ratio
TSA	trial sequential analysis
VKA	vitamin K antagonist

stroke prevention for patients with AF who are unable to tolerate long-term anticoagulation.

The initial randomized trial (PROTECT AF) showed that LAAC was noninferior to VKA for stroke prevention and was superior for cardiovascular and all-cause mortality during long-term follow-up.⁶ However, in the PREVAIL trial, LAAC did not meet the prespecified noninferiority criteria for stroke, systemic embolism, and cardiovascular death compared with VKA at the primary as well as 5-year analysis.⁶ The PRAGUE-17 trial showed noninferiority of

LAAC to DOACs for a composite safety and efficacy end point that also included clinically relevant bleeding.⁷ The combination of safety and efficacy end points is controversial, especially when the results appear to be driven by clinically relevant bleeding. In patients with AF after catheter ablation, LAAC was superior to OAC in reducing nonprocedure major bleeding at 3 years but was noninferior when procedural bleeding was included.⁸ LAAC also showed noninferiority to OAC in a composite end point of death, stroke, or systemic embolism using a prespecified 5% noninferiority margin.⁸ Concurrently, early rhythm control alone has demonstrated a significantly lower incidence and absolute event rates of stroke than usual care.⁹

The CLOSURE-AF trial renewed scrutiny of LAAC by not demonstrating noninferiority to best medical therapy (primarily DOACs) for a composite end point that included stroke in patients at high risk of both stroke and bleeding.¹⁰ However, the recently published CHAMPION-AF trial compared LAAC with DOACs and showed noninferiority of LAAC for the composite end point of cardiovascular death, stroke, or systemic embolism and superiority for nonprocedural bleeding.¹¹ Notably, these trials were primarily powered for composite end points, which may favor noninferiority in the presence of competing risks and obscure differences in individual outcomes, particularly ischemic stroke and procedural bleeding, highlighting an important gap in the current evidence. In this study, we conducted an updated systematic review and meta-analysis focusing on individual outcomes to define the comparative risk-benefit profile of LAAC relative to OAC. We also performed a trial sequential analysis (TSA) to determine whether the cumulative evidence reached the required information size and to evaluate the robustness of the findings.

METHODS

Data Availability

The authors declare that all supporting data are available within the article and its Supplemental Material.

Ethics Statement

This study was a systematic review and meta-analysis of published RCTs that did not involve individual patient-level data. Institutional review board approval and informed consent were not required. The study was registered in PROSPERO (URL: <https://www.crd.york.ac.uk/PROSPERO/>; Unique identifier: CRD420261367479). The Preferred Reporting Items for Systematic Reviews and Meta-Analyses checklist and the Grading of Recommendations Assessment, Development, and Evaluation assessment for the study are provided in Tables S1 and S2.²¹

Search Strategy, Study Selection, and Eligibility Criteria

The systematic review and meta-analysis were performed in accordance with the Preferred Reporting Items for Systematic

Reviews and Meta-Analyses guideline.¹² We systematically searched the MEDLINE (via PubMed), Cochrane Central Register of Controlled Trials, Embase, Cumulative Index of Nursing and Allied Health, and ClinicalTrials.gov databases from January 1, 2000 to March 31, 2026 for relevant published articles comparing LAAC devices with OAC in patients with AF. The search used the following keywords: (Atrial fibrillation) AND (Left atrial appendage closure OR Left atrial appendage occlusion OR Watchman device OR Amulet device) AND (anticoagulation OR direct oral anticoagulation OR vitamin K antagonist). We also reviewed the references in the retrieved articles, relevant reviews, and prior meta-analyses for additional published data.

We included studies enrolling adults (≥ 18 years) with AF that compared LAAC with OAC and reported outcomes, including mortality, stroke, systemic embolism, and major bleeding. We excluded abstracts, case reports, data derived solely from review articles, observational or registry studies, and non-English publications. Two authors (V.R. and R.C.) independently screened articles for inclusion, retrieved potentially relevant studies, and determined eligibility. Disagreements were resolved by consensus or consultation with a third author (A.R.). The study selection process was documented using a Preferred Reporting Items for Systematic Reviews and Meta-Analyses flow diagram (Figure 1).

Data Extraction and Quality Assessment

Using a standardized protocol, all relevant information from each article was extracted by 3 authors (V.R., R.C., and A.R.), and information on study and patient characteristics, intervention details, and outcomes were collected and summarized. Risk of bias was independently assessed by 2 investigators (A.R. and R.C.) using the Cochrane Risk of Bias 2 tool for randomized controlled trials (RCTs).¹³ Two authors (A.R. and R.C.) independently assessed the study quality.

Statistical Analysis

Meta-analyses were performed conservatively using an a priori random-effects model with the Mantel-Haenszel method, assuming substantial variability in the treatment effect size across trials.¹⁴ Principal summary effect measures were expressed as risk ratios (RRs) with 95% CIs. The primary efficacy end point was any stroke, and the primary safety end point was major bleeding, including procedural and nonprocedural bleeding. For each outcome, the number of events and the total number of patients in each treatment arm were extracted and pooled. Prespecified subgroup analyses were performed, including comparisons between VKAs and DOACs.

Sensitivity analyses were performed sequentially by excluding individual studies to assess the robustness of the findings and to identify which studies influenced the overall effect size. Statistical heterogeneity was evaluated using the Cochran Q test and quantified using the I^2 statistic; when I^2 exceeded 50%, study-level characteristics were explored, and leave-one-out sensitivity analyses were performed.¹⁵ Publication bias was assessed using funnel plots and the Egger test.¹⁶

Statistical analyses were performed using Review Manager (RevMan, version 5.3; Copenhagen: The Nordic Cochrane Centre, The Cochrane Collaboration, 2014), and Stata 18 (StataCorp LLC, College Station, TX).^{17,18}

Statistical significance was assumed at a 2-sided $P < 0.05$. TSA was performed to further assess the robustness of the pooled estimates using the TSA software (version 0.9.5.10 Beta [Copenhagen Trial Unit, Rigshospitalet, Copenhagen, Denmark]).^{19,20} A 2-sided type I error of 5% and a power of 80% were prespecified for all outcomes, with anticipated relative risk reductions of 20%. Control event rates were derived from the OAC groups of the included RCTs.

RESULTS

Our meta-analysis included 6 RCTs comparing LAAC with OAC in patients with AF (PROTECT AF, PREVAIL, PRAGUE-17, OPTION, CHAMPION-AF, and CLOSURE-AF), as shown in Figures 2 and 3. We also included the 5-year outcomes data from the PROTECT AF and PREVAIL studies.^{6-8,10,11,22,23} The primary pooled analysis comprised 7004 patients (3681 LAAC; 3323 OAC).

The baseline characteristics showed a mean age of 71.6 to 78.5 years, with a male preponderance across all trials (Tables 1 and 2). Stroke risk profiles were substantial, with mean CHADS₂ or CHA₂DS₂-VASC scores of 2.3 in PROTECT AF, 3.9 in PREVAIL, 4.7 in PRAGUE-17, 3.5 in OPTION, 5.2 in CLOSURE-AF, and 3.5 in CHAMPION-AF.^{7,8,10,11,22,23} Mean Hypertension, Abnormal Renal/Liver Function, Stroke, Bleeding History or Pre-disposition, Labile International Normalized Ratio, Elderly, Drugs/Alcohol (HAS-BLED) scores were relatively low at ≈ 1 , except in CLOSURE-AF and PRAGUE-17, in which mean HAS-BLED scores were > 3 .^{7,8,10,11,22,23}

LAAC was associated with a significantly higher risk of ischemic stroke compared with OAC (RR, 1.41 [95% CI, 1.04–1.91]; $P = 0.03$; 3.2% versus 2.1%; $n = 3480$ versus $n = 3122$). There was no significant heterogeneity ($I^2 = 0\%$). However, the benefit appeared to be driven by the recent results from CHAMPION-AF.¹¹ There was no significant difference in the risk of any stroke (RR, 1.08 [95% CI, 0.82–1.42]; $P = 0.60$; 4.0% versus 3.4%; $n = 3681$ versus $n = 3323$) or in the composite end point of stroke, systemic embolism, or death (RR, 0.99 [95% CI, 0.77–1.29]; $P = 0.96$; 6.6% versus 5.9%; $n = 3034$ versus $n = 2680$). Cardiovascular mortality was also similar between groups (RR, 0.88 [95% CI, 0.64–1.19]; $P = 0.40$; 5.7% versus 5.8%; $n = 3681$ versus $n = 3323$), as was all-cause mortality (RR, 0.92 [95% CI, 0.78–1.10]; $P = 0.36$; 10.9% versus 11.1%; $n = 3681$ versus $n = 3323$).

Bleeding outcomes were largely comparable, with no significant difference in major bleeding, including procedural bleeding (RR, 0.93 [95% CI, 0.80–1.10]; $P = 0.40$; 7.9% versus 8.1%; $n = 3681$ versus $n = 3323$). Hemorrhagic stroke was numerically lower with LAAC but was not statistically significant (RR, 0.56 [95% CI, 0.28–1.11]; $P = 0.10$; 0.7% versus 1.1%; $n = 3480$ versus $n = 3122$). In contrast, LAAC was associated with a significantly lower risk of nonprocedural major bleeding (RR, 0.57 [95% CI, 0.43–0.77]; $P < 0.01$; 6.2% versus 10.7%; $n = 3235$ versus $n = 2881$). $I^2 < 50\%$ was observed among

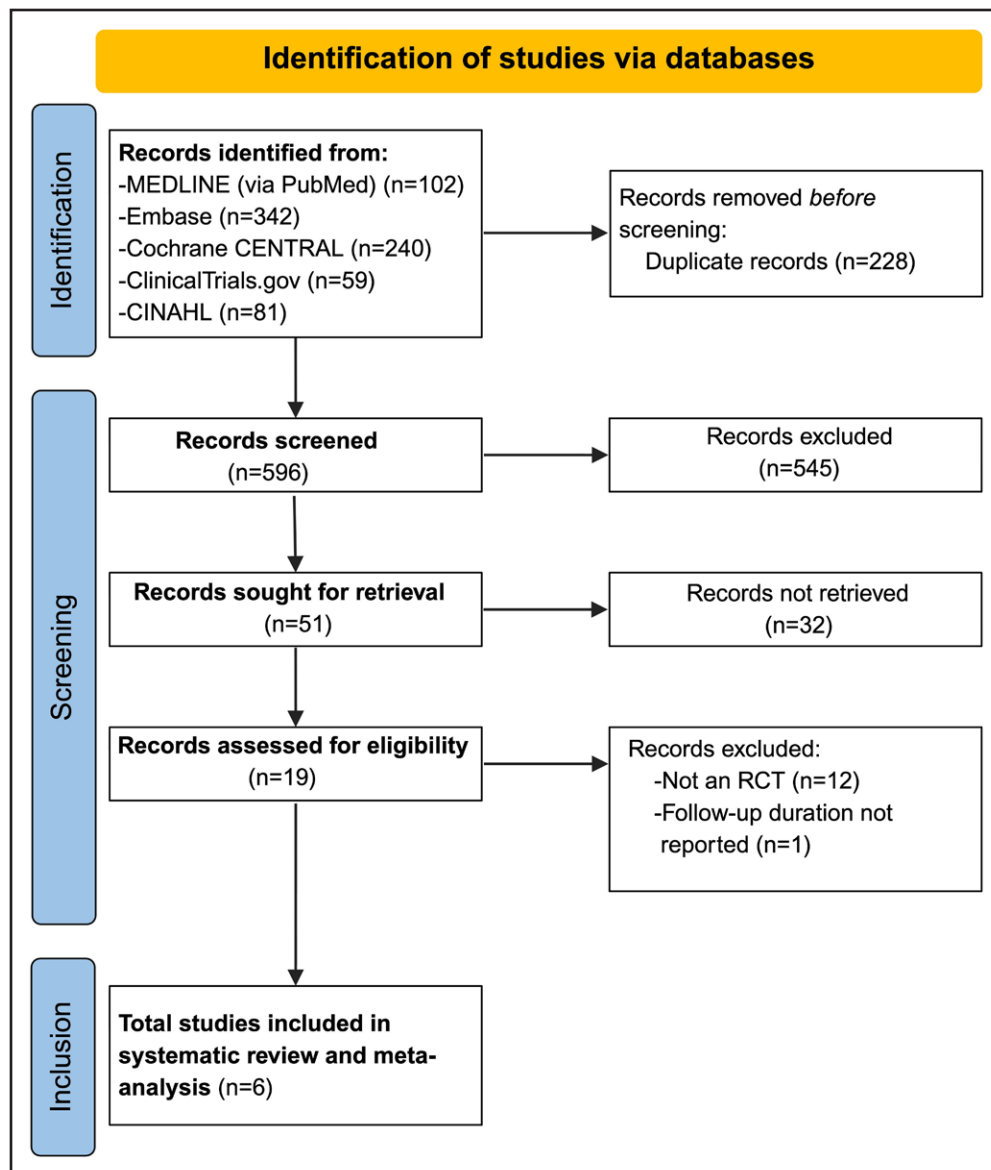


Figure 1. Study selection flow diagram according to Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines.

PRISMA flow diagram of study selection. A total of 824 records were identified, 596 were screened, and 6 randomized controlled trials were included after screening. CENTRAL indicates Cochrane Central Register of Controlled Trials; and CINAHL, Cumulative Index of Nursing and Allied Health. Created in BioRender. Rao, A. (2026) <https://BioRender.com/vqfk8vy>

trials for all outcomes except cardiovascular mortality and nonprocedural major bleeding. Leave-one-out sensitivity analyses for cardiovascular mortality and major bleeding showed similar results after sequential exclusion of individual studies (Figures S1 and S2).

TSA reached the required statistical information size for all outcomes except hemorrhagic stroke. Statistical information size was used instead of sample size to account for low event rates (Figures S3 through S10). For ischemic stroke, TSA demonstrated the benefit of OAC over LAAC, although the proximity of the Z curve to the upper monitoring boundary limits the robustness of this finding and warrants cautious interpretation. For any

stroke, cardiovascular mortality, major bleeding (including procedural), and the composite end point, the cumulative Z curve showed no clinically meaningful effect at the prespecified threshold, supporting the primary pooled analysis. In contrast, TSA supported a robust benefit in nonprocedural major bleeding with LAAC.

Subgroup Analysis: Type of Anticoagulation

Outcomes from the subgroup analysis were broadly consistent with the overall analysis, with some notable differences (Figures 4 and 5). Unlike the overall analysis, LAAC, compared with VKAs, showed no significant

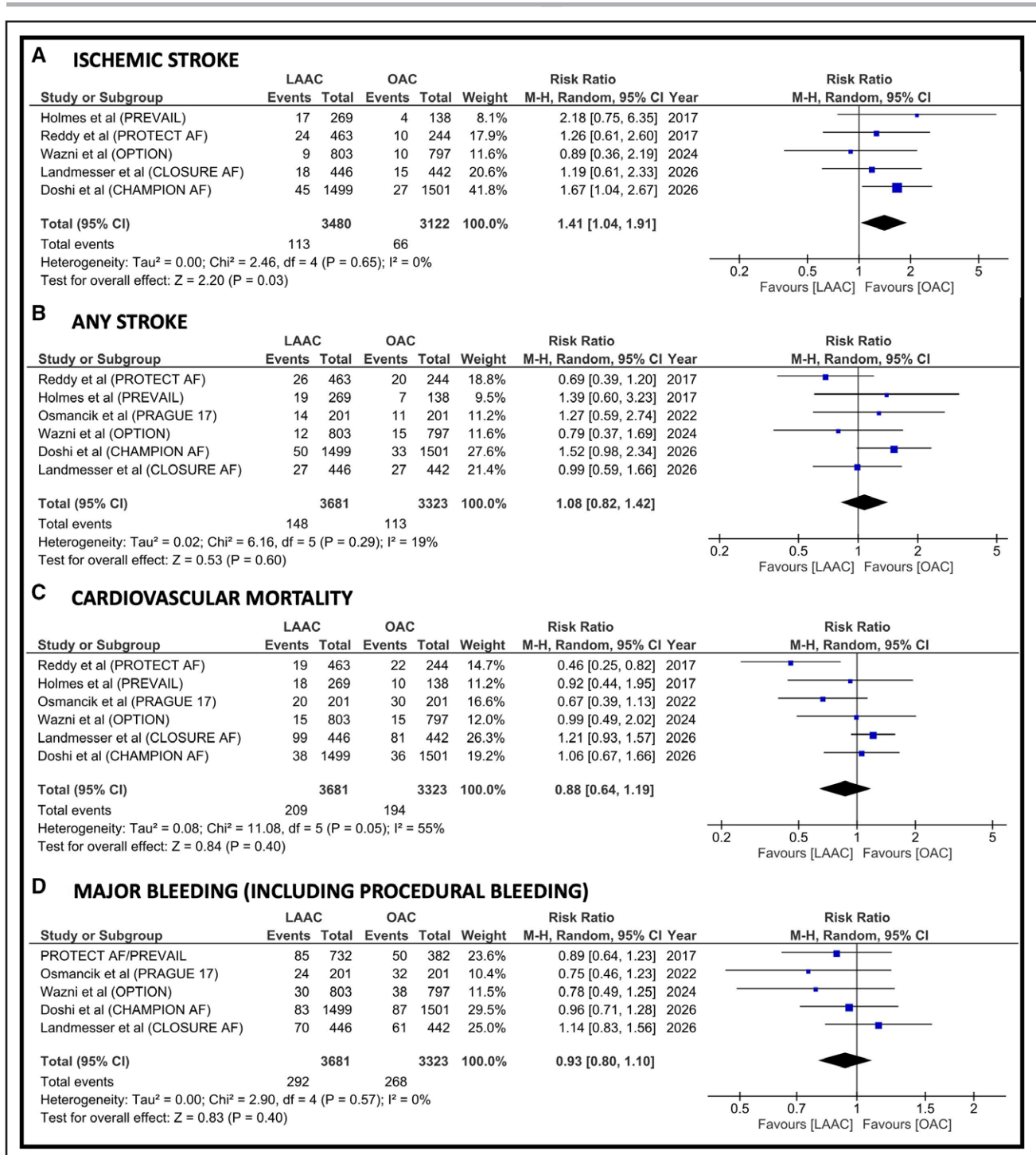


Figure 2. Comparative effects of left atrial appendage closure (LAAC) vs oral anticoagulation (OAC) on ischemic stroke, overall stroke burden, cardiovascular mortality, and major bleeding (including procedural bleeding).

Forest plots comparing LAAC with OAC for outcomes. Risk ratios with 95% CIs were calculated using a Mantel-Haenszel random-effects model. Blue squares represent study-specific estimates weighted by inverse variance, and black diamonds represent pooled effect estimates. Values <1 favor LAAC.

increase in the risk of ischemic stroke (RR, 1.50; $P=0.18$). However, LAAC did demonstrate lower risk of all-cause mortality (RR, 0.76 [95% CI, 0.58–0.99]; $P=0.05$), hemorrhagic stroke (RR, 0.21 [95% CI, 0.07–0.58]; $P<0.01$), and nonprocedural major bleeding (RR, 0.49 [95% CI,

0.34–0.71]; $P<0.001$), compared with VKAs. There was no significant difference in any stroke (RR, 0.91; $P=0.78$), cardiovascular mortality (RR, 0.83; $P=0.77$), major bleeding, including procedural bleeding (RR, 0.89; $P=0.47$), and the composite outcome (RR, 0.82; $P=0.25$).

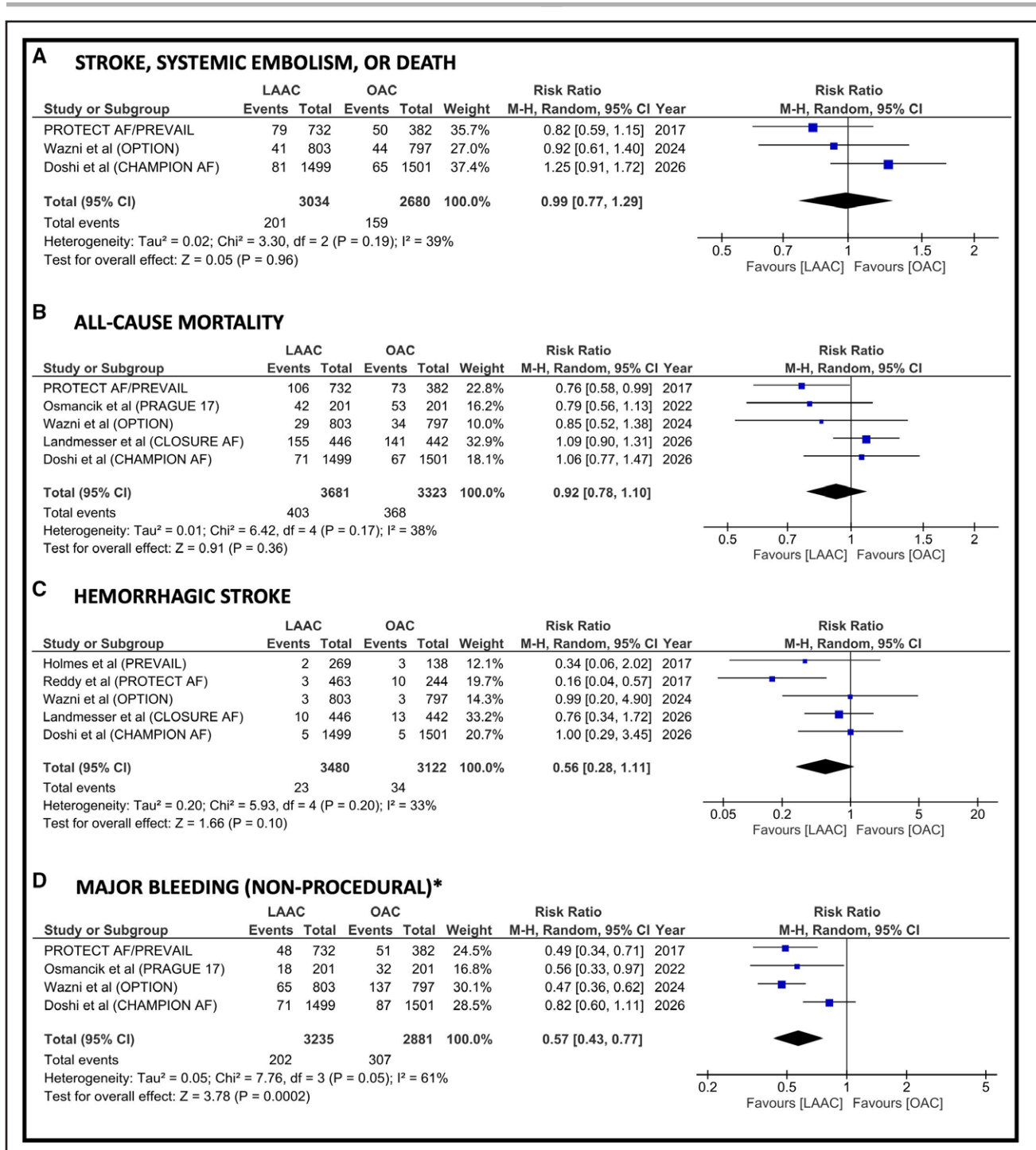


Figure 3. Comparative effects of left atrial appendage closure (LAAC) vs oral anticoagulation (OAC) on composite thromboembolic outcomes, all-cause mortality, hemorrhagic stroke, and nonprocedural bleeding.

Forest plots comparing LAAC with OAC for outcomes. Risk ratios with 95% CIs were calculated using a Mantel-Haenszel random-effects model. Blue squares represent study-specific estimates weighted by inverse variance, and black diamonds represent pooled effect estimates. Values <1 favor LAAC. *PRAGUE-17 did not provide nonprocedural major bleeding, so nonprocedural clinically relevant bleeding data were used.⁷

When compared with DOACs, LAAC was not associated with an increased risk of ischemic stroke (RR, 1.38; $P=0.08$). There was no significant difference in any stroke (RR, 1.19; $P=0.23$), cardiovascular mortality (RR, 0.78; $P=0.36$), major bleeding, including procedural bleeding (RR, 0.95; $P=0.58$), composite outcome (RR,

1.11; $P=0.49$), all-cause mortality (RR, 1.01; $P=0.91$), or hemorrhagic stroke (RR, 0.85; $P=0.61$). LAAC demonstrated lower risk of nonprocedural major bleeding (RR, 0.60 [95% CI, 0.41–0.88]; $P=0.03$). Similar to the overall analysis, $I^2<50\%$ was observed among trials included in the subgroup analysis for all outcomes

Table 1. Design of the Randomized Controlled Trials Included in the Analysis

Study	Funding; location; follow-up duration	Inclusion criteria	Exclusion criteria	Type of LAAC	Type of AC	Primary efficacy end point	Primary safety end point
1. Reddy et al ^{6,22} (PROTECT AF); 2014 and 2017	Boston Scientific; United States+Europe (59 sites); 5 y	NVAF, CHADS ₂ ≥1	PFO with ASA, ASD, Mechanical valve, contraindication to warfarin, LVEF <30%	Watchman 2.5*	Warfarin	Composite: stroke, SE, CV/unexplained death	Procedure-related complications (pericardial effusion, hospitalization, device embolization). Major bleeding
2. Holmes et al ^{6,23} (PREVAIL); 2014 and 2017	Boston Scientific; Primarily United States, multicenter; 5 y	NVAF, CHADS ₂ ≥2 OR ≥1+high-risk features	Recent stroke (<90 d), need for long-term OAC other than AF, PFO/ASD, carotid disease, clopidogrel indication	Watchman 2.5*	Warfarin	1. Composite: stroke, SE, CV/unexplained death 2. Late ischemic stroke or SE >7 d	Early safety composite: all-cause death, ischemic stroke, SE, or major procedural complication within 7 d or during index hospitalization
3. Osmancik et al ⁸ (PRAGUE-17); 2022	Investigator-initiated; Czech Republic (10 sites); 4 y	NVAF with high stroke or bleeding risk (CHA ₂ DS ₂ -VASc ≥3+HAS-BLED ≥2 OR prior bleeding/cardioembolism on AC)	Mechanical valve, MS, CrCl <30 mL/min, LAA thrombus, severe renal dysfunction, PFO/ASA, recent bleeding/stroke within 30 d, non-AF OAC indication	Watchman*/Watchman FLX* or Amulet†	DOAC (mostly apixaban)	Composite: stroke/TIA, systemic embolism, CV death, clinically relevant bleeding, procedural/device complications	Included in composite (net clinical benefit design)
4. Wazni et al ⁹ (OPTION); 2025	Boston Scientific; International (10 countries, 106 sites); 3 y	NVAF postablation (90–180 d prior or planned within 10 d), CHA ₂ DS ₂ -VASc ≥2 (men)/≥3 (women)	Mechanical valve, MS, LVEF <30%, ASD/PFO or repair, stroke within 60 d, bleeding within 14 d, non-AF OAC indication	Watchman FLX*	DOAC (physician choice)	Composite: all-cause death, stroke, or systemic embolism at 36 mo	Nonprocedure-related major or clinically relevant nonmajor bleeding
5. Landmesser et al ¹⁰ (CLOSURE-AF); 2026	DZHK (Germany); multicenter (42 sites); 3 y (1.7–4.7)	NVAF and CHA ₂ DS ₂ -VASc ≥2+HAS-BLED score of ≥3/other high bleed risk/contraindication to long-term OAC	Mechanical valve, GFR <15, liver failure, heart transplant, symptomatic carotid disease, non-AF OAC indication	Watchman*/Watchman FLX*/Amulet† (CE devices)	Best medical therapy (mostly DOAC±SAPT/DAPT)	Composite: CV or unexplained death, stroke, SE, major bleeding (net clinical benefit variants)	Included in composite (net clinical benefit design). Procedural bleeding included.
6. Doshi et al ¹¹ (CHAMPION-AF); 2026	Boston Scientific; International (16 countries, 141 sites); 3 y	NVAF, CHA ₂ DS ₂ -VASc ≥2 (men) ≥3 (women), eligible for DOAC therapy	Mechanical valve, stroke/TIA/major bleeding within 30 d, PFO/ASA, LVEF <30%, non-AF OAC indication, MI	Watchman FLX*	DOAC	Composite: CV death, stroke, SE at 36 mo	Nonprocedural ISTH major+clinically relevant nonmajor bleeding at 36 mo

AF indicates atrial fibrillation; ASA, atrial septal aneurysm; ASD, atrial septal defect; CE, Conformité Européenne; CrCl, creatine clearance; CV, cardiovascular; DAPT, dual antiplatelet therapy; DOAC, direct oral anticoagulation; GFR, glomerular filtration rate; HAS-BLED, Hypertension, Abnormal Renal/Liver Function, Stroke, Bleeding History or Predisposition, Labile International Normalized Ratio, Elderly, Drugs/Alcohol; ISTH, International Society on Thrombosis and Hemostasis; LAA, left atrial appendage; LAAC, LAA closure; LVEF, left ventricular ejection fraction; MI, myocardial infarction; MS, mitral stenosis; NVAF, nonvalvular atrial fibrillation; OAC, oral anticoagulation; PFO, patent foramen ovale; SAPT, single antiplatelet therapy; SE, systemic embolism; and TIA, transient ischemic attack.

*Boston Scientific Inc.

†Abbott Inc.

except cardiovascular mortality and major or clinically relevant nonprocedural bleeding.

Risk of Bias and Grading of Recommendations Assessment, Development, and Evaluation

Risk of bias was low across most domains, although PRAGUE-17 had a higher risk related to crossover and deviations from intended interventions (Figure S11).⁷

Grading of Recommendations Assessment, Development, and Evaluation assessment showed moderate certainty for ischemic stroke and major bleeding (including procedural bleeding), low certainty for any stroke, composite outcome, all-cause mortality, hemorrhagic stroke, and nonprocedural major bleeding, and very low certainty for cardiovascular mortality (Table S2). Visual inspection of funnel plots did not show marked asymmetry, and the Egger test was

Table 2. Baseline Characteristics of Randomized Controlled Trials Included in the Analysis Comparing LAAC Versus OAC Groups

Study	Reddy et al ^{6,22} (PROTECT AF); 2014 and 2017	Holmes et al ^{6,23} (PREVAIL); 2014 and 2017	Osmancik et al ⁷ (PRAGUE-17); 2022	Wazni et al ⁸ (OPTION); 2025	Landmesser et al ¹⁰ (CLOSURE-AF); 2026	Doshi et al ¹¹ (CHAMPION-AF); 2026
Total no. (LAAC vs OAC)	463 vs 244	269 vs 138	201 vs 201	803 vs 797	446 vs 442	1499 vs 1501
Age, y/male, % (LAAC vs OAC)	71.7±8.8 vs 72.7±9.2/70.4% vs 70.1%	74.0±7.4 vs 74.9±7.2/67.7% vs 74.6%	73.4±6.7 vs 73.2±7.2/66.7% vs 64.7%	69.7±7.4 vs 69.4±7.9/64.8% vs 66.9%	78.5±6.8 vs 77.3±7.3/61.4% vs 61.3%	71.6±7.5 vs 71.8±7.5/67.6% vs 68.4%
HTN/DM/CKD (LAAC vs OAC)	89.6% vs 90.2%/24.4% vs 29.5%/NR	88.5% vs 97.1%/33.8% vs 29.7%/ NR	92.5% vs 92.5%/36.3% vs 44.8%/NR	89.8% vs 88.0%/27.6% vs 27.7%/NR	93.5% vs 94.3%/39.2% vs 42.1%/25.6% vs 22.6%	NR
CHA ₂ DS ₂ -VASc score/HAS-BLED score (LAAC vs OAC)	2.2±1.2 vs 2.3±1.2*/NR	3.8±1.2 vs 3.9±1.2/NR	4.7±1.5 vs 4.7±1.5/3.1±0.9 vs 3.0±0.9	3.5±1.3 vs 3.5±1.3/1.2±0.8 vs 1.2±0.8	5.2±1.5 vs 5.1±1.6/3.1±0.9 vs 3±0.9	3.5±1.2 vs 3.5±1.3/1.3±0.8 vs 1.3±0.8
Type of AF (LAAC vs OAC)	Paroxysmal 43.2% vs 40.6%; persistent 21.0% vs 20.5%; permanent 34.6% vs 38.1%	Paroxysmal 48.7% vs 51.4%; persistent 31.6% vs 28.3%; permanent 15.6% vs 15.9%	Paroxysmal 26.4% vs 33.3%; persistent 23.4% vs 22.9%; long- standing persistent 9.0% vs 8.0%; permanent 41.3% vs 35.8%	Paroxysmal 59.4% vs 62.9%; persistent 40.6% vs 37.1%	Paroxysmal 68.7% vs 68.9%; persistent 27.9% vs 24.2%; permanent 3.4% vs 6.9%	Paroxysmal 69.3% vs 68.5%; persistent 23.9% vs 25.4%; permanent 6.84% vs 6.1%
AF ablation (LAAC vs OAC)	NR	NR	NR	100% vs 100%	NR	48.7% vs 46.9%
LVEF/LA size (LAAC vs OAC)	57.3±9.7% vs 56.7±10.1%/NR	55.4±10% vs 56.0±9.8%/NR	52.9±12.1% vs 53.3±12.6%; 51.7±7.5 mm vs 51.8±6.4 mm	NR	NR	NR

AF indicates atrial fibrillation; CKD, chronic kidney disease; DM, diabetes; HAS-BLED, Hypertension, Abnormal Renal/Liver Function, Stroke, Bleeding History or Predisposition, Labile International Normalized Ratio, Elderly, Drugs/Alcohol; HTN, hypertension; LA, left atrium; LAAC, left atrial appendage closure; LVEF, left ventricular ejection fraction; NR, not reported; and OAC, oral anticoagulation.

*CHADS₂ score.

not statistically significant across all outcomes (all $P>0.05$; Figures S12 through S19).

DISCUSSION

In this systematic review and meta-analysis of 6 RCTs enrolling 7004 patients, LAAC was compared with OAC for stroke prevention in patients with AF. We report 3 principal findings: (1) LAAC was associated with a higher risk of ischemic stroke; (2) there was no significant difference in any stroke or major bleeding, including procedural bleeding; and (3) the VKA subgroup showed higher rates of all-cause mortality, hemorrhagic stroke, and non-procedural major bleeding compared with LAAC.

Our analysis builds on prior findings from individual RCTs, which suggested a trend toward a higher risk of ischemic stroke with LAAC but were underpowered to evaluate this further. We found that LAAC demonstrated a 41% increase in risk of ischemic stroke, with an absolute risk increase of 1.1% over the 3- to 5-year follow-up period. Across the trials in our analysis, definitions of ischemic stroke were generally consistent and included periprocedural events, although granular details on disabling and nondisabling strokes remain unavailable.^{7,8,10,11,22} Periprocedural events, thromboembolic sources outside the left atrial appendage, peri-device

leak, and device-related thrombus formation may have contributed to the increased risk of ischemic stroke.²⁴ On the contrary, LAAC showed a trend for lower risk of hemorrhagic stroke compared with OAC. When LAAC was compared with the VKA group alone, it led to a 79% lower risk of hemorrhagic stroke, with an absolute risk reduction of 2.7%. However, the risk of hemorrhagic stroke remained similar between the LAAC and DOAC groups. There was no significant difference between LAAC and OAC for the outcome of any stroke. Although results from TSA do not favor robustness in increased ischemic stroke, this information needs to be taken into strong consideration for shared decision-making, given the volume of LAAC devices being implanted, until future trials and long-term follow-up provide more clarity.

In our analysis, there were no significant differences in individual outcomes, including all-cause mortality, cardiovascular mortality, and any stroke, between LAAC and OAC. This is consistent with the noninferiority signal of LAAC relative to OAC for the primary efficacy end point in most individual trials. However, it is important to note that PREVAIL did not meet the noninferiority criteria for the primary efficacy end point even at long-term follow-up. CLOSURE-AF, which included bleeding outcomes in the composite end point, also showed that LAAC was not noninferior to physician-directed care, which mostly

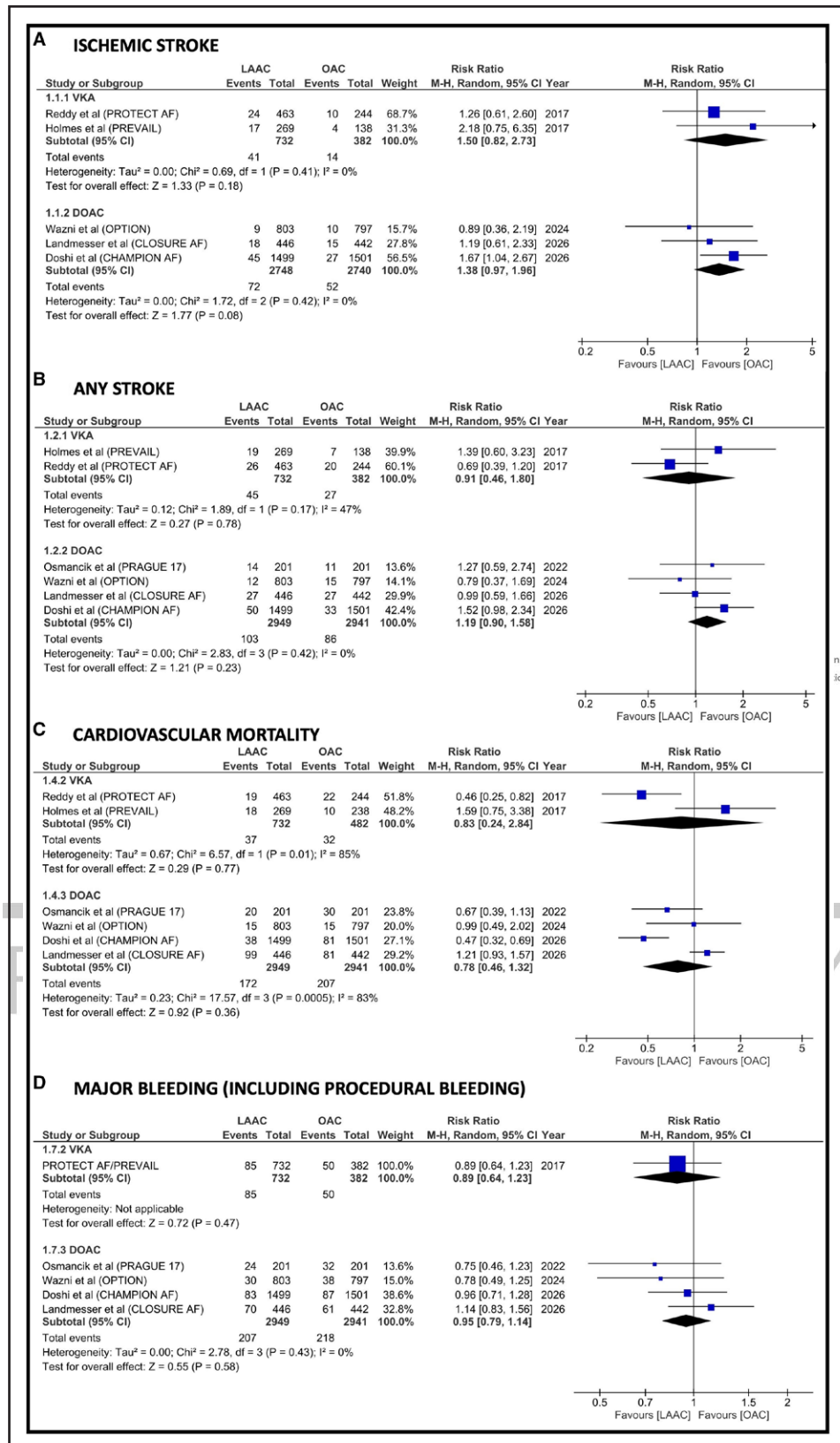


Figure 4. Comparative effects of left atrial appendage closure (LAAC) vs oral anticoagulation (OAC) on ischemic stroke, overall stroke burden, cardiovascular mortality, and procedural bleeding between vitamin K antagonist (VKA) and direct OAC (DOAC) subgroups.

Forest plots comparing LAAC with OAC between VKA and DOAC subgroups for outcomes. Risk ratios with 95% CIs were calculated using a Mantel-Haenszel random-effects model. Blue squares represent study-specific estimates weighted by inverse variance, and black diamonds represent pooled effect estimates. Values <1 favor LAAC.

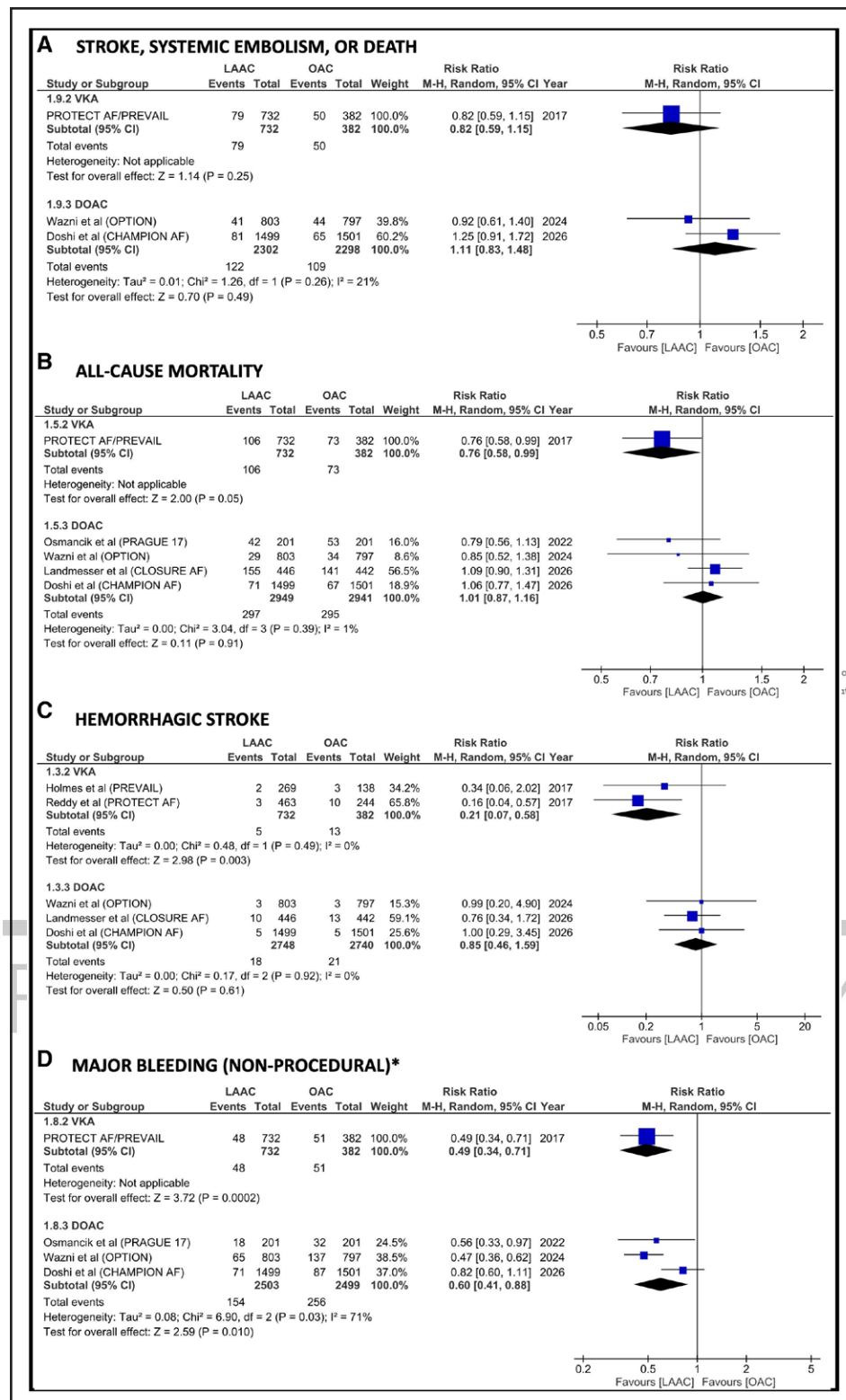


Figure 5. Comparative effects of left atrial appendage closure (LAAC) vs oral anticoagulation (OAC) on composite thromboembolic outcomes, all-cause mortality, hemorrhagic stroke, and non-procedural bleeding between vitamin K antagonist (VKA) and direct OAC (DOAC) subgroups.

Forest plots comparing LAAC with OAC between VKA and DOAC subgroups for outcomes. Risk ratios with 95% CIs were calculated using a Mantel-Haenszel random-effects model. Blue squares represent study-specific estimates weighted by inverse variance, and black diamonds represent pooled effect estimates. Values <1 favor LAAC. *PRAGUE-17 did not provide nonprocedural major bleeding, so nonprocedural clinically relevant bleeding data were used.⁷

included DOACs.¹⁰ This raises concern about whether the use of LAAC in patients at high bleeding risk may demonstrate unfavorable outcomes and needs to be evaluated in future trials. Overall, the results of these individual outcomes were corroborated by our TSA, which demonstrated that the required information size was reached. It must be noted that variance-based statistical information was used in our TSA instead of the conventional sample size, given the low event rate. Although methodologically appropriate, it is inherently sensitive to assumptions and should be interpreted with caution. This underscores the need for larger, adequately powered trials to confirm the findings. In the subgroup analysis, LAAC was associated with a lower risk of all-cause mortality compared with VKAs, suggesting that LAAC might be the preferred approach for patients who remain on VKAs due to financial constraints or who experience adverse effects from DOACs.

When procedural bleeding was included, there was no significant difference in major bleeding between groups. Although the trials used major bleeding definitions from different classification systems (Table S3), the definitions generally captured clinically significant hemoglobin decline, need for transfusion, or bleeding at critical sites.^{7,8,10,11} Some trials additionally defined clinically relevant nonmajor bleeding (CRNMB) and included it either as a separate end point (OPTION and CHAMPION-AF) or a composite end point (PRAGUE-17).^{7,8,11} Three trials (PROTECT AF, PREVAIL, CLOSURE-AF) did not include CRNMB, and LAAC was notably not noninferior to OAC in CLOSURE-AF, even though the primary outcome included bleeding.^{10,22,23} There was also variation in the definition of CRNMB across trials. Although PRAGUE-17 defined CRNMB as requiring hospitalization or invasive intervention, OPTION defined it as bleeding requiring medical intervention, hospitalization, or an increased level of care, prompting face-to-face evaluation.^{7,8} CHAMPION-AF defined CRNMB as bleeding requiring any medical intervention or hospitalization.¹¹ Although hospitalization or invasive intervention are important considerations, other definitions could introduce bias, especially because CRNMB appears to be a driver of trials that favor LAAC.

Our study demonstrated a 43% lower risk of nonprocedural major bleeding with LAAC compared with OAC. Both the CHAMPION-AF and PRAGUE-17 trials showed reductions in clinically relevant nonprocedural bleeding with LAAC.^{7,11} The exclusion of procedural bleeding is a criticism of the OPTION, CHAMPION-AF, and PRAGUE-17 trials, which potentially overestimate the benefit of LAAC compared with OAC.^{7,8,11} Disregarding periprocedural events, such as bleeding, is generally discouraged, as these events are associated with greater long-term mortality and morbidity in patients undergoing cardiovascular procedures.^{25,26} However, procedural and nonprocedural bleeding likely represent distinct clinical

entities. Procedure-related bleeding is often mechanical in origin, occurs in a monitored setting, and is generally recognized and managed promptly. In contrast, nonprocedural bleeding may reflect a patient's underlying comorbidity burden and ongoing antithrombotic exposure and often occurs during long-term follow-up. Consistent with this distinction, several studies have reported a stronger association between non-access-site or spontaneous bleeding and subsequent mortality compared with procedure-related access-site bleeding, suggesting that not all bleeding events carry the same prognostic implications.^{27,28} In the 4-year follow-up of the PRAGUE-17 trial, clinically relevant nonprocedural bleeding showed progressive curve separation favoring LAAC, with the benefit most apparent after the first year, indicating that its benefit may emerge with longer follow-up as bleeding curves progressively separate.⁷ This may eventually overcome the upfront procedural bleeding outcomes. Studies with long-term follow-up >3 years would therefore be required to better demonstrate this difference.

It is worth noting that the CLOSURE-AF trial included older and sicker patients, including those with higher CHA₂DS₂-VASc and HAS-BLED scores than those in other similar trials.¹⁰ In addition, >40% of patients received the Amulet device (Abbott), introducing heterogeneity in device use.¹⁰ The seemingly neutral results of CLOSURE-AF may therefore reflect a higher baseline risk with competing mortality and increased periprocedural risks of LAAC negating long-term benefit.¹⁰ The physician-directed care group in CLOSURE-AF predominantly used DOACs in this high-bleeding-risk group, yet did not demonstrate a higher bleeding risk.¹⁰ This raises a question of whether practitioners in the real-world underuse DOACs among patients with prior bleeding, and whether a more proactive approach to DOAC use in these patients might be justified over LAAC.²⁹ However, we have to be pragmatic about whether this approach is sustainable outside of the realm of a rigorous randomized trial.

Two recent trials suggest discontinuing DOAC might be an option for intermediate-risk postablation patients with AF.^{30,31} The ALONE-AF trial (n=840) demonstrated that in a moderate-risk cohort (median CHA₂DS₂-VASc score of 2; 69.5% with scores ≤2; 67.6% with paroxysmal AF), stopping DOAC was associated with a lower risk of the composite of stroke, systemic embolism, and major bleeding at 2 years.³⁰ Similarly, in the larger OCEAN trial (n=1284; mean CHA₂DS₂-VASc score of 2.2; 68.1% with scores ≤2; 66.4% with paroxysmal AF), DOAC did not differ from aspirin in the composite outcome of stroke, systemic embolism, or covert embolic stroke at 3 years.³¹ However, it was associated with a higher risk of minor bleeding, CRNMB, and the composite of major and minor bleeding.³¹ Hence, discontinuation of DOACs after ablation may be reasonable in these patient populations. Importantly, these populations differed substantially

from those enrolled in the LAAC trials included in our analysis, which generally enrolled older patients with higher stroke risk and a greater proportion of persistent or long-standing persistent AF. In addition, ALONE-AF and OCEAN had relatively short follow-up durations of 2 to 3 years.^{30,31} As stroke risk evolves, longer follow-up may reveal differences that are not apparent during early follow-up. The progressive separation of nonprocedural bleeding curves observed in the PRAGUE-17 trial suggests that bleeding-related benefits may become more pronounced over time.⁷ In addition, in patients with $\text{CHA}_2\text{DS}_2\text{-VASc} \geq 3$ and at high bleeding risk, offering concomitant LAAC placement during catheter ablation may further reduce procedural bleeding by avoiding a second procedure. However, this approach needs to be confirmed in future trials.

The current meta-analysis expands on the scope of previous work by incorporating 6 RCTs with a total of 7004 patients, compared with 4 RCTs and 3116 patients evaluated in a recent analysis.³² Although the previous meta-analysis concluded that LAAC was associated with significant reductions in all-cause and cardiovascular mortality, this updated analysis, which included the recently published CHAMPION-AF and CLOSURE-AF trials, found no significant difference in mortality outcomes between LAAC and OAC.^{10,11,32} A key difference lies in the efficacy findings: the previous study reported no significant difference in ischemic events, whereas our study identifies a statistically significant 41% higher risk of ischemic stroke associated with LAAC.³² Furthermore, by using TSA, our study provides a higher level of statistical rigor than the previous study.³² Finally, by focusing on individual clinical outcomes, this analysis addresses the limitations of composite end points that can obscure differences in treatment effects.

Overall, despite the seemingly favorable results of the CHAMPION-AF trial, there remains a significant reason to exercise caution in the use of LAAC.¹¹ The question regarding the most appropriate candidate for LAAC is multifaceted. The only RCT that included patients at high bleeding risk (HAS-BLED score >3 , prior bleeding precluding anticoagulation) showed that LAAC was not noninferior to OAC.¹⁰ Intuitively, this would be the group assumed to benefit from LAAC. Patients with a $\text{CHA}_2\text{DS}_2\text{-VASc}$ score <2 derive more benefit from successful rhythm control and discontinuation of OAC.^{30,31} Patients with a $\text{CHA}_2\text{DS}_2\text{-VASc}$ score ≥ 3 and a HAS-BLED score ≤ 2 , as represented in CHAMPION-AF, or where rhythm control is challenging, could be considered for LAAC, while recognizing that the risk of ischemic stroke is higher, whereas the risks of any stroke and major bleeding are not significantly different.¹¹ In addition, shared decision-making about the relevance of CRNMB to patients and understanding of their financial constraints should also be emphasized.

Our study indicates a higher risk of ischemic stroke with LAAC compared with OAC, whereas the risks of any stroke, hemorrhagic stroke, composite outcomes, and both cardiovascular and all-cause mortality are similar. The risk of major bleeding when including procedural events was comparable, but excluding procedural bleeding showed a reduced risk with LAAC. Our subgroup analysis also found more favorable outcomes with LAAC than with VKAs, whereas no significant differences were observed compared with DOAC therapy.^{10,11}

Study Limitations

Several limitations should be acknowledged. First, heterogeneity in comparator arms across trials (VKAs in PROTECT AF/PREVALE versus DOACs/best medical therapy in PRAGUE-17, OPTION, CHAMPION-AF, and CLOSURE-AF) may influence pooled estimates.^{7,8,10,11,22,23} DOACs have superior efficacy and safety compared with warfarin, and the inclusion of warfarin-controlled trials may bias the results in favor of LAAC. Second, the substantial heterogeneity observed for some end points (cardiovascular mortality, $I^2=55\%$; nonprocedural bleeding, $I^2=61\%$) suggests variability in treatment effects across populations. Third, a majority of patients in the included trials had moderate stroke risk ($\text{CHA}_2\text{DS}_2\text{-VASc}$ 3–4), and results may not be generalizable to patients with very high stroke risk ($\text{CHA}_2\text{DS}_2\text{-VASc} \geq 6$), who were largely underrepresented. Similarly, patients at high bleeding risk were also underrepresented. Fourth, most trials used the Watchman devices (Watchman/Watchman FLX; Boston Scientific), limiting extrapolation to other LAAC devices. Finally, this analysis was limited to published aggregate data rather than individual patient-level data, which prevents detailed subgroup analyses and time-to-event modeling.

Future Directions

Several ongoing and planned studies will further inform the role of LAAC in AF management.^{33,34} Longer-term follow-up of the CHAMPION-AF and CLOSURE-AF trials will provide additional data on the durability of treatment effects.^{10,11} In addition, advances in imaging, including intracardiac echocardiography and computed tomography, may improve patient selection and procedural planning. Furthermore, the ongoing LAAOS-4 trial (The Fourth Left Atrial Appendage Occlusion Study) is evaluating whether percutaneous LAAC provides incremental benefit when added to continued OAC, rather than serving as a substitute for anticoagulation.³⁵ By comparing LAAC plus OAC with OAC alone, this study will help determine whether a combined mechanical and pharmacological strategy can further reduce thromboembolic risk in patients with AF and may identify patient populations most likely to benefit from adjunctive LAAC.³⁵

Finally, the integration of LAAC with catheter ablation for AF, as evaluated in the OPTION trial, represents an emerging paradigm that may expand the population of patients who benefit from LAAC.⁸

Conclusions

In this meta-analysis of 6 RCTs enrolling 7004 patients, LAAC was associated with a higher risk of ischemic stroke, but a significantly lower risk of nonprocedural bleeding compared with OAC. No significant differences were observed in the composite of stroke, systemic embolism, or death, or in any stroke, hemorrhagic stroke, all-cause or cardiovascular mortality, and major bleeding when including procedural bleeding. These findings support caution in the use of LAAC as an alternative to anticoagulation for stroke prevention in AF, with treatment selection guided by individual patient risk profiles and patient preferences.

ARTICLE INFORMATION

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Affiliations

Department of Internal Medicine, Guthrie Robert Packer Hospital, Sayre, PA (A.R.). Division of Electrophysiology, Heart and Vascular Institute, Saint Francis Health System, Tulsa, OK (V.R.). Division of Cardiology, Cook County Health, Chicago, IL (M.M., A.V.). Department of Cardiology, University of Miami Miller School of Medicine, FL (R.C.). Vydehi Institute of Medical Sciences and Research Centre, Bengaluru, India (V.T.). Jagadguru Jayadeva Murugarajendra Medical College, Davanagere, India (N.N.). Division of Cardiology, Rush University Medical Center, Chicago, IL (G.D., N.J., A.V.).

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Supplemental Material

Tables S1–S3
Figures S1–S19

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